

BenMedHallu

A Fact-Controlled Benchmark for Hallucination Detection in Bengali Medical Text

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Four task formats · **17,915** labelled items · **82,622** detector judgements ·
six LLMs

Introduction

A **hallucination** in clinical text generation is **unfaithfulness to a source**, not implausibility. It has two forms: **extrinsic** – the output asserts something the source never stated – and **intrinsic** – it asserts something the source contradicts.

In Bengali medical text both are **invisible at the surface**. Each of these edits leaves grammatical, register-consistent, confident clinical prose:

হিস্টাসিন	→	সিটিরিজিন	same drug class, different drug
৫.৯ ইঞ্চি	→	৫.৯ ফুট	unit rescaled
রেজাল্ট পজিটিভ	→	রেজাল্ট ভালো	specific value generalised away

ROUGE and BERTScore are near-invariant to exactly these edits, and the reader most exposed to the error is least equipped to catch it.

Problem statement. No resource measures whether LLMs can distinguish faithful Bengali medical text from deliberately corrupted text, and no methodology exists for building one in which the **gold label**, the **error type** and the **item difficulty** are all under the builder's control.

Introduction – Objectives and Contributions

Four challenges specific to this problem:

Distribution leakage. If hallucinated items are machine-written and faithful ones human-written, a detector wins by recognising generated prose.

Label certainty. An edit meant to remove support may fail silently, mislabelling the item and penalising every detector that answers correctly.

Register preservation. Patients repeat details in several forms, so a partial edit leaves the document self-contradicting – a different phenomenon from the one intended.

Attributable difficulty. For a score to be diagnostic rather than merely low, only one experimental axis may move at a time.

Contributions:

- The first hallucination-detection benchmark for Bengali medical text – **four task formats, 17,915 labelled items, 82,622 judgements.**
- A construction method that **derives gold labels rather than annotating them**, recording the changed span with every item.
- **Coarsen**, a novel corruption operation isolating *partial* support – the hardest condition in the benchmark.
- An evaluation of six detectors disaggregated by entity type, corruption operation and cost.

Related Works – BenHalluEval

ADIB, SANI, ESHAM, ABRAR, TASHDEED & CHOWDHURY (2026) – IUT & UC

Problem. No prior work systematically evaluated hallucination in LLMs for Bengali, the sixth most spoken language.

Solution. A four-task framework (QA, code-mixed QA, summarization, mathematical reasoning); 12,000 hallucinated candidates generated with GPT-5.4; nine LLMs evaluated under a **dual-track protocol**.

Results. BenHalluScore 7.72–55.42%; three native annotators, Fleiss' $\kappa = 0.911$ –0.926; CoT prompting helps on 18 of 32 combinations and hurts on 14.

What we adopt

The Track A / Track B protocol, and the BanglaCHQ-Summ corpus.

What we invert

BenHalluEval corrupts the **summary** (Fabricated Content, Non-Factual Addition, Direct Contradiction).

BenMedHallu corrupts the **source** and never edits the human summary.

Establishes the dual-track baseline for Bengali; BenMedHallu carries it into the clinical domain.

Related Works – Fact-Controlled Diagnosis (Leave- N -Out)

SUHAS BN ET AL., INTERSPEECH 2025

Problem. Benchmarks built by annotating model output cannot say whether detection degrades for one corruption type over another – type was never a controlled variable.

Solution. Extract atomic facts from a reference, remove N of them from the input, and treat N as an explicit **difficulty dial** rather than a post-hoc bin. Applied to English clinical dialogue (ACI-Bench).

Impact. Establishes construction-based labelling for clinical summarization, with difficulty controlled by design.

Closest ancestor of this work

We adopt Leave- N -Out directly, with $N \in \{1, 2, 3\}$.

Two differences

ACI-Bench holds *structured English* clinical encounters; our corpus holds *unstructured Bengali* patient messages.

We add two alteration operations alongside deletion.

Related Works – MedHallu

PANDIT ET AL., EMNLP 2025

Problem. General-domain hallucination benchmarks understate the risk in medicine, where an error is clinically consequential rather than merely wrong.

Solution. 10,000 medical QA items built on PubMedQA – **1,000 human-annotated and 9,000 artificially generated** – with difficulty-graded hallucinations.

Results. Domain specificity amplifies the impact of hallucination; supplying context before the judgement, and offering a third “Not Sure” option, both improve detection.

Methodological sibling

The same choice we make: **manufacture** the hallucinations rather than wait for a model to produce them.

Gap it leaves

English, journal-derived text, and hallucinations confined to medical terminology – not the informal, code-mixed patient register we target.

Related Works – Medical hallucination beyond English

MED-HALT – PAL ET AL., CONLL 2023

Problem. Hallucination testing had not reached the medical domain.

Solution. 59k items derived from medical **examinations** (PubMed, MedMCQA, MedQA-USMLE) across English, Spanish and Chinese, with reasoning and memory-based tests.

Limitation. Frames hallucination as *incorrect recall* rather than unfaithfulness to a source; remains English-centric.

Direct ancestor of our Tracks 2 and 3 – exam items, and an abstention option.

CMHE – CHINESE MEDICAL HALLUCINATION EVAL, LREC-COLING 2024

Problem. Medical hallucination evaluation existed only for English.

Solution. 42k items from Chinese medical dialogue data, with a **detection / diagnosis / explanation** benchmark; a portion re-verified by medical professionals.

Result. Establishes that a dedicated, expert-validated medical hallucination benchmark is feasible outside English.

The gap, stated by someone else: this exists for Chinese. It does not exist for Bengali.

Related Works – Positioning and Identified Gaps

Work	Task	Language	Label provenance
TruthfulQA	QA truthfulness	English	Human curation
HaluEval	QA, dialogue, summ.	English	Sampling + human filter
FactCC	Summarization consistency	English	Rule-based synthesis
FActScore	Atomic factual precision	English	Retrieval + LM validation
SelfCheckGPT	Detection method	English	Sampling consistency
Med-HALT	Medical reasoning	English-centric	Exam-derived
MedHallu	Medical QA	English	1k human + 9k generated
CMHE	Medical detection	Chinese	Generated + expert re-check
Leave- <i>N</i> -Out	Clinical summarization	English	By construction
BenHalluEval	Four tasks	Bengali	Generated + 3 annotators
BanglaSummEval	Summarization eval.	Bengali	LLM judges
BenMedHallu	Four formats, medical	Bengali	By construction

Linguistic gap – medical hallucination benchmarks exist for English and Chinese; the Bengali entries are general-domain. **Label-provenance gap** – annotation-based benchmarks label the errors they encountered, not the errors they designed. **Circularity gap** – where generator, detector and judge share a family, agreement inside that family is indistinguishable from faithfulness.

Methodology – The Inversion

Conventional construction

Keep the source fixed, ask a model to rewrite the **output**, and label the rewrite as hallucinated.

Consequence: every hallucinated item is machine-written and every faithful item is human-written. The two classes differ in **fluency, length and idiom** as well as in faithfulness.

A detector can then score well by recognising **generated prose** – without ever checking a fact.

BenMedHallu

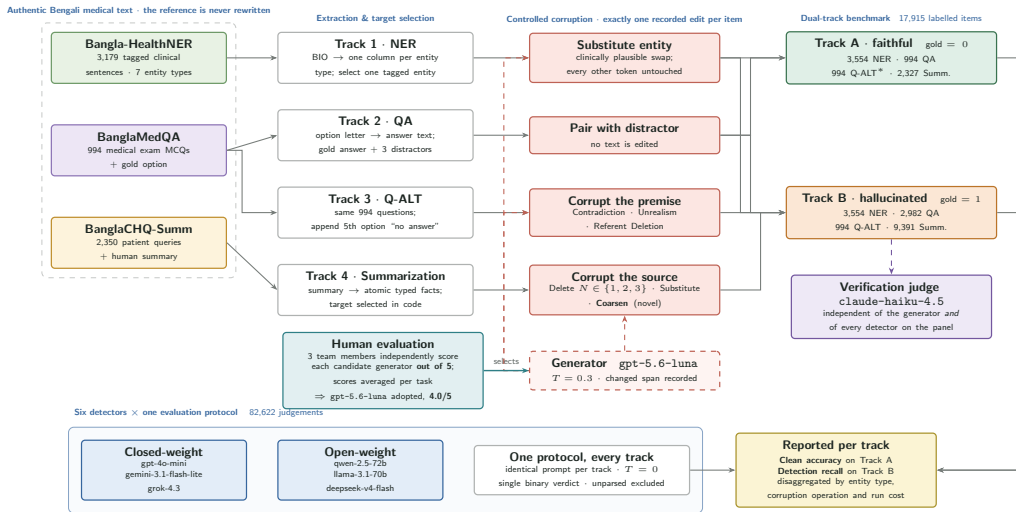
Take authentic Bengali medical text known to be faithful and apply exactly **one controlled, recorded edit** that destroys that faithfulness.

The reference is **never rewritten**, so faithful and hallucinated items share one text distribution.

The label is an **output of the pipeline** rather than a judgement about it, and the record of what changed comes free.

The same inversion runs through all four tracks. **NER** corrupts the sentence around a tagged entity, not the entity label. **QA** generates no text at all – it pairs an untouched question with its real exam distractors. **Q-ALT** corrupts the *question*, not the answer. **Summarization** corrupts the *source*, not the summary.

Methodology – System Pipeline



Methodology – Construction: Tracks 1 and 2

Track 1 – Entity-level (NER)

Extraction. BIO tagging converted to one column per entity type, giving every sentence its tagged clinical entities.

Selection. One tagged entity per sentence, chosen in code.

Generation. The generator replaces it under four constraints: the replacement must be **clinically plausible, different** from the original, **absent elsewhere** in the sentence, and **every other token untouched**.

Evaluation. The detector is shown the sentence and the target entity and returns one binary digit. **Each row is judged twice** – clean and corrupted.

Track 2 – Answer-level (QA)

Extraction. The gold option letter is resolved to its full answer text.

Selection. Each question is paired once with the correct answer and once with each of its **three real exam distractors**.

Generation. **None.** No text is corrupted – the distractors were written by the examiners.

Evaluation. The detector judges whether the proposed answer is factually correct for the question.

Why three variants. They agree within about three points, which shows the ranking is not an artefact of which distractor was drawn.

Methodology – Construction: Tracks 3 and 4

Track 3 – Question-level (Q-ALT)

The same 994 questions, with a fifth option “উত্তর নেই” (*no answer*) appended. Each question is rewritten so that **no option A–D can be correct**, making E the only valid choice. Three techniques:

Contradiction – two individually plausible but jointly impossible premises.

Unrealism – a clinically styled but factually wrong value or property.

Referent Deletion – removes the detail behind a confident anaphoric reference whose antecedent was never supplied.

Only selecting the fifth option counts as detection.

Track 4 – Document-level (Summarization)

Stage 1 – Extraction. The human summary is decomposed into **atomic typed facts** across ten clinical categories.

Stage 2 – Selection. A **seeded RNG selects the target in code**, not the model. This preserves the gold record, the category balance and the difficulty dial, and makes reruns reproducible.

Stage 3 – Generation. The source is rewritten so the target loses support while every retained fact survives. Three operations: **Delete** $N \in \{1, 2, 3\}$, **Substitute**, **Coarsen**.

Stage 4 – Verification. An independent judge checks each fact before acceptance.

Three support relations, by design. **Delete** leaves the claim *unsupported* (extrinsic). **Substitute** leaves it *contradicted* (intrinsic). **Coarsen** replaces a specific value with a vaguer one still true of the original – *partial* support.

Methodology – Selection and Protocol

Generator selection was measured, not assumed.

A pilot study scored the candidates two ways: **three team members independently rated each generator's output out of five** per task, averaged below; and an automated bake-off scored the same items with independent judges, where a *lower* detection rate indicates harder items.

Candidate generator	Medicine	Symptom	Health Cond.	Summarization	Mean
gpt-5.6-luna	4.0	4.0	4.0	4.0	4.00
deepseek-v4-pro	4.0	4.0	3.5	3.0	3.63
claude-opus-5	3.0	4.0	–	–	3.50
qwen-3.8-max	3.5	3.0	–	–	3.25

Mean human rating out of 5 over three independent reviewers. gpt-5.6-luna was best or tied on every task and was adopted as the sole generator.

One protocol for every track. An identical prompt per track is issued to all six detectors at $T = 0$; each returns a single binary verdict; unparseable verdicts are **excluded from the denominator** rather than defaulted.

Anti-circularity. Generator, detectors and judge come from **disjoint model families**; a panel member is never the judge, so family agreement cannot be mistaken for faithfulness.

Implementation

Source corpora

Bangla-HealthNER	Khan et al., <i>NERvous About My Health</i> , EMNLP Findings 2023 – 3,179 tagged clinical sentences
BanglaCHQ-Summ	Khan et al., ACL 2023 Architecture – 2,350 patient queries with human summaries
bangla-med-qa	994 medical examination MCQs with gold options

A four-stage repository: `preprocessing/` holds the generation pipelines, `dataset/` the assembled benchmark, `evaluation_*/` the detector runs, and aggregation scripts produce the reported tables. Prompts and selection logic live in **importable modules**, **not notebooks**, so each prompt exists in exactly one place.

Models – all eight reached through OpenRouter, giving one client, one retry policy and one cost ledger across six vendors.

Generator	gpt-5.6-luna	Reliability at
Judge	claude-haiku-4.5	
Detectors	gpt-4o-mini	
	gemini-3.1-flash-lite	
	grok-4.3	
	qwen-2.5-72b-instruct	
	llama-3.1-70b-instruct	
	deepseek-v4-flash	

scale

Runs against six commercial APIs at tens of thousands of calls fail constantly. Results are **checkpointed every fifty rows** and every run resumes from its own output file; multiple API keys rotate on HTTP 402 and 429; a repair pass re-runs only rows whose verdict is null; and a **pre-flight probe** issues one call per detector before committing a full run.

Results – Benchmark Inventory and Track 1

Benchmark inventory

Track	Items	Judg.
NER	3,554	38,540
QA	3,976	23,856
Q-ALT	994	5,964
Summarization	9,391	14,262
Total	17,915	82,622

Judgement counts reflect *actual* detector coverage rather than the nominal six models.

Clean accuracy – accuracy on the faithful member of a pair.

Recall – proportion of corrupted items correctly flagged.

The gap between them is the principal finding.

Track 1 – detection recall (%) by entity type

Detector	Med.	Age	Dos.	H.C.	Spec.	Proc.
Grok-4.3	94.7	–	–	69.2	83.5	73.6
DeepSeek V4 Flash	79.6	96.6	83.0	68.5	80.2	51.2
Gemini 3.1 fl.-lite	75.3	–	87.3	57.0	64.9	62.0
GPT-4o-mini	73.3	94.7	82.2	61.7	64.6	53.7
Qwen 2.5 72B	69.9	99.6	81.0	67.9	72.3	55.8
Llama 3.1 70B	54.4	73.1	–	22.7	15.9	31.8

A dash marks a detector not run on that category.

Entity type strongly predicts difficulty. Age and Dosage are close to solved (73–99.6%): the corrupted value is a *number*, and a number mismatching its context is easy to spot. Health Condition, Specialist and Medical Procedure require *clinical knowledge* rather than pattern matching, and every detector falls away.

The Llama row. At 15.9% on Specialist it accepts five of six swapped entities while scoring 93.9% on the matching clean sentences – agreeing with the prompt rather than reading the pair.

Results – Tracks 2 and 3

Track 2 – answer-level (%)

Detector	Accepts correct	Rejects wrong
Grok-4.3	86.3	92.2
Gemini 3.1 flash-lite	86.1	90.0
DeepSeek V4 Flash	81.0	89.6
GPT-4o-mini	66.9	84.7
Qwen 2.5 72B	67.2	78.2
Llama 3.1 70B	71.1	70.8

Wrong-answer recall is the mean over three distractor variants, which agree within about three points.

The bias direction reverses relative to Track 1. There, models *under*-flag and accept corrupted text. Here they *over*-flag: GPT-4o-mini rejects a third of the medically correct answers while still catching 84.7% of the wrong ones. One model carries two opposite failure modes depending on framing – the clearest evidence for building four task formats rather than one.

Track 3 – question-level, detection rate (%)

Detector	Contra.	Unreal.	Ref. Del.	Overall
Grok-4.3	95.0	91.9	82.6	88.8
DeepSeek V4 Flash	72.0	76.7	47.0	60.6
Gemini 3.1 fl.-lite	62.4	62.8	46.4	54.8
GPT-4o-mini	46.1	58.1	53.8	50.8
Qwen 2.5 72B	51.8	47.7	37.9	44.9

Items: Contradiction 436, Referent Deletion 472, Unrealism 86.
Llama 3.1 70B is excluded – its run stalled after 211 of 994 items.

The hardest track in the benchmark, and technique matters as much as model. **Referent Deletion** is hardest for four of five detectors, and by a wide margin – DeepSeek drops from 72.0% to 47.0%.

The reason is structural: Contradiction and Unrealism both insert something *false*, which clinical knowledge can catch. Referent Deletion inserts nothing false at all – it *removes* something, leaving a confident reference with no antecedent.

Results – Track 4 and the Operation Gradient

Track 4 – overall (%) and run cost

Detector	Cln.	Cor.	Ovr.	USD
Grok-4.3	89.1	94.2	93.2	4.567
Gemini 3.1 fl.-lite	89.3	93.5	92.6	0.137
Qwen 2.5 72B	89.6	87.8	88.1	0.489
GPT-4o-mini	93.6	78.0	81.1	0.098
Llama 3.1 70B	78.0	81.1	80.5	0.697
DeepSeek V4 Flash	89.6	72.9	76.2	0.121

Scale and price are weak predictors. Grok-4.3 finishes first at 93.2% but costs **\$4.57** against **\$0.137** for Gemini 3.1 flash-lite, which finishes **0.58 points behind** – a thirty-three-fold cost difference for a gap well inside single-run noise.

GPT-4o-mini again inverts its Track 1 profile: best of all six on untouched pairs at 93.6%, but only 78.0% on corrupted sources.

Recall (%) by corruption operation

Detector	Del.1	Del.2	Del.3	Subst.	Coars.
Grok-4.3	98.0	100.0	100.0	95.9	72.7
Gemini 3.1 fl.-lite	97.4	100.0	100.0	94.0	71.7
Qwen 2.5 72B	90.6	99.2	99.7	84.3	62.3
Llama 3.1 70B	85.3	92.1	96.0	72.2	59.3
DeepSeek V4 Flash	70.9	82.8	84.8	73.1	51.2
GPT-4o-mini	77.6	91.3	93.3	80.2	42.7
Mean	86.6	94.2	95.6	83.3	60.0

Two clean gradients.

(1) Deletion becomes easier as N rises, from 86.6% at one fact to 95.6% at three. More missing material means more signal, and the trend holds across *all six* detectors – confirming N works as the intended difficulty dial.

(2) Coarsen collapses. At a mean of 60.0% it sits 23 points below Substitute and 36 below three-fact deletion; GPT-4o-mini catches only 42.7%. The source still discusses the detail and points the same way, so a detector checking *topical agreement* finds it and stops.

Results – Qualitative Samples Across All Four Tracks

Bold figures give how many detectors correctly flagged the item. Samples span the difficulty range.

Track 1 – NER

Medicine হিস্টাসিন → সিটিরিজিন **3/6**
Same drug class, different drug.

Specialist “Heart er doctors” → “Pulmonologist” **3/6**
Cardiologist to pulmonologist.

Health Cond. বদহজম → গ্যাস্ট্রাইটিস **3/6**
An adjacent diagnosis, not the same one.

Track 2 – QA

মাথার খুলিতে কয়টি হাড় রয়েছে? **3/6**
Proposed answer **23**; three of six accepted it.

Track 3 – Q-ALT

Contradiction **2/5**
The deltoid tuberosity placed on *both* the scapula and the humerus – anatomically impossible.

Track 4 – Summarization

Coarsen **0/6**
রেজাল্ট পজিটিভ আসছে → রেজাল্ট ভালো আসছে
summary claims প্রোগনোসিস টেস্ট পজিটিভ
“Came out good” no longer supports “positive”.

Substitute **0/6**
শ্বাসকষ্ট হয় কিছুটা → মাথা ঘোরে কিছুটা
summary claims শ্বাসকষ্ট হয় কিছুটা
Breathing difficulty became dizziness.

Delete $N=1$ **2/6**
87 characters removed, including the whole account of the 14-day antibiotic course healing the fissure – yet the summary still claims it healed.

Challenges and Limitations

Challenges

Inconsistent verdict polarity. The NER prompts were developed in two groups with opposite conventions – for some categories a verdict of 1 means *supported*, for others *hallucinated*. Applying one convention uniformly silently inverts half the results rather than producing an obvious error.

API reliability at scale. Credit exhaustion and rate limiting interrupted long runs constantly. Key rotation, checkpointing and repair passes recovered most but not all: the Llama Q-ALT run stalled at 211 of 994 items.

Reasoning models returning empty completions. A model whose reasoning budget exceeds the token cap returns an empty string rather than an error – discovered only after a batch had been spent, which led to the mandatory pre-flight probe.

Cost control. One frontier detector cost \$4.57 on a single track against \$0.10–\$0.14 for flash-tier models, forcing a decision about which tracks the expensive models could cover.

Limitations

Verification not applied to the released set. The judge is implemented but was not run over the assembled dataset within the timeline, so released items passed only the two mechanical filters. A residual proportion may carry an incorrect gold label, which would depress all six detectors' recall roughly equally – the reported **rankings are therefore more trustworthy than the absolute values**.

Coverage. Symptom, the largest entity type, reached generation only. Age and Dosage were evaluated by four detectors rather than six. The Llama Q-ALT run is incomplete, and 7,483 corrupted summarization items in the training split remain unscored.

No clinical validation. All corruptions are model-generated; no clinician has confirmed the substituted entities are plausible in context.

Statistical treatment. A single prompt at temperature 0, with no prompt-sensitivity study or significance testing. Differences of one or two points should not be treated as meaningful

Conclusion and Future Direction

The central result is a gap, not an average. Six contemporary models read faithful Bengali medical text almost perfectly and miss between 5% and 84% of the corruptions inserted into it. That gap widens systematically as the corruption becomes *less like an error and more like an omission*.

Findings against the research questions

RQ1 – affirmatively but narrowly: frontier models can detect Bengali medical hallucination, but only Grok-4.3 does so reliably on *both* halves of a pair.

RQ2 – entity type strongly predicts difficulty: numeric types reach 73–99.6%, types requiring clinical knowledge fall to 15.9–86.4%.

RQ3 – the operation matters more than the entity type, with mean recall spanning 36 points from Coarsen (60.0%) to three-fact deletion (95.6%).

RQ4 – negatively: a 70B open-weight model places last on every track it completed, and a flash-tier model matches a frontier reasoning model within 0.6 points at one thirty-third the cost.

Future work

1. Run the verification judge. Stage 4 is written and parked; executing it converts a well-filtered dataset into a verified one.

2. Close the coverage gaps. Score the Symptom track, add the missing detectors to Age and Dosage, rerun Llama, and score the unevaluated summarization split.

3. Score span localisation, which needs no new data – every item already stores the original and replacement spans.

4. Human clinical validation of a stratified sample by Bangladeshi clinicians.

5. Cheap supervised baselines. With the benchmark in place, fine-tuning BanglaBERT tests whether a small supervised model can approach a commercial API.

6. Other low-resource languages. The pipeline is language-agnostic apart from its prompts.

Coarsening a value rather than falsifying it defeats every detector tested, and no existing benchmark isolates that condition.

GitHub and Individual Contribution

Repository <https://github.com/nashatislam/benmedhallu-priv>

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• Track 1 (NER) generation and evaluation pipelines • Track 2 (QA) construction and detector runs • Repository architecture and cost ledger • Human evaluation reviewer	• Track 4 (Summarization) source-corruption pipeline • Fact extraction and Coarsen operation design • Verification judge implementation • Human evaluation reviewer	• Track 3 (Q-ALT) technique design and generation • NER category expansion and evaluation • Results aggregation and reporting tables • Human evaluation reviewer

Literature survey, benchmark design decisions, pilot generator selection and the final report were carried out jointly.

Thank You